

LAB EXPERIMENT RECORD-29/09/2026**Set 1: Monthly Sales Data**

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Tuesday

Aim

To visualize monthly sales data using a line chart, bar chart, and scatter plot, and to build an interactive dashboard using R and Shiny.

Dataset: Monthly Sales

Month	Sales (\$)
January	15000
February	18000
March	22000
April	20000
May	23000

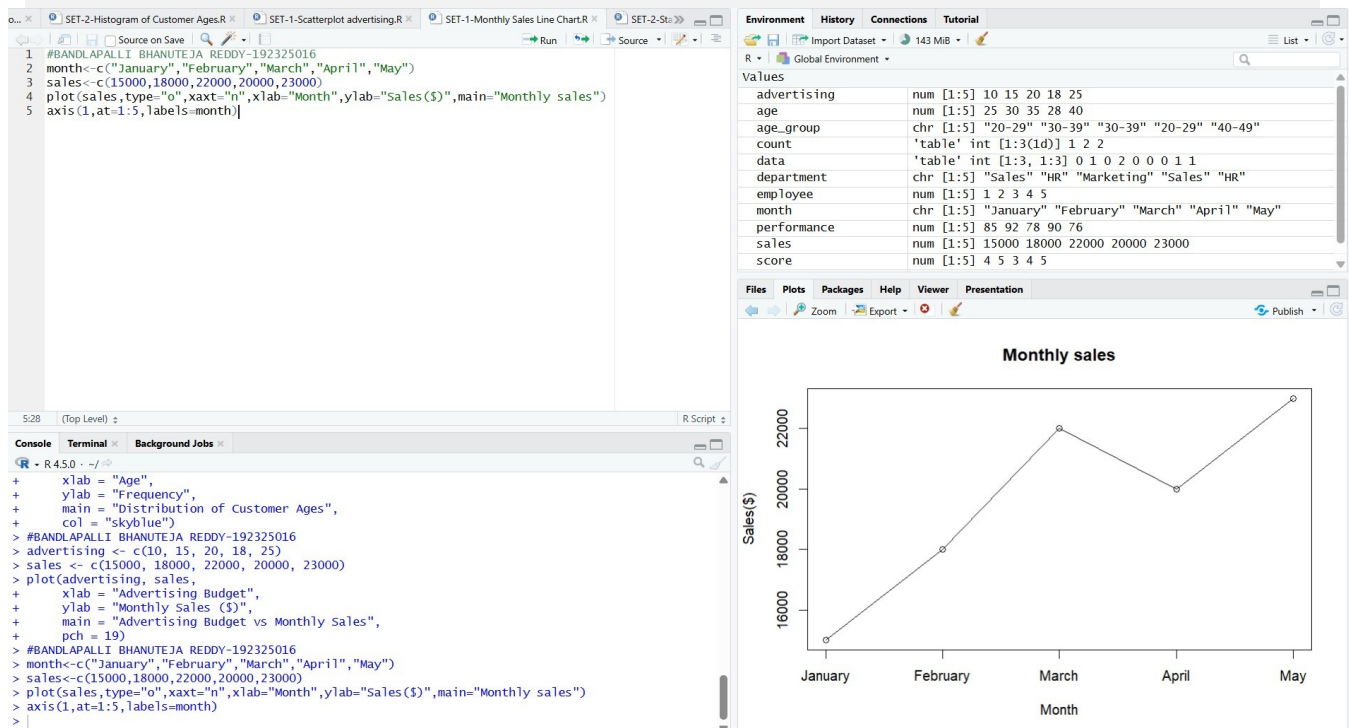
PROGRAM 1: Line Chart of Monthly Sales

Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

```
# Dataset month <- c("January", "February", "March",
"April", "May") sales <- c(15000, 18000, 22000, 20000,
23000)
```

```
# Line Chart plot(sales,
type = "o", xaxt = "n",
xlab = "Month", ylab =
"Sales ($)", main =
"Monthly Sales", col =
"blue")
```

```
axis(1, at = 1:5, labels = month)
```



Output 1: Monthly Sales Line Chart

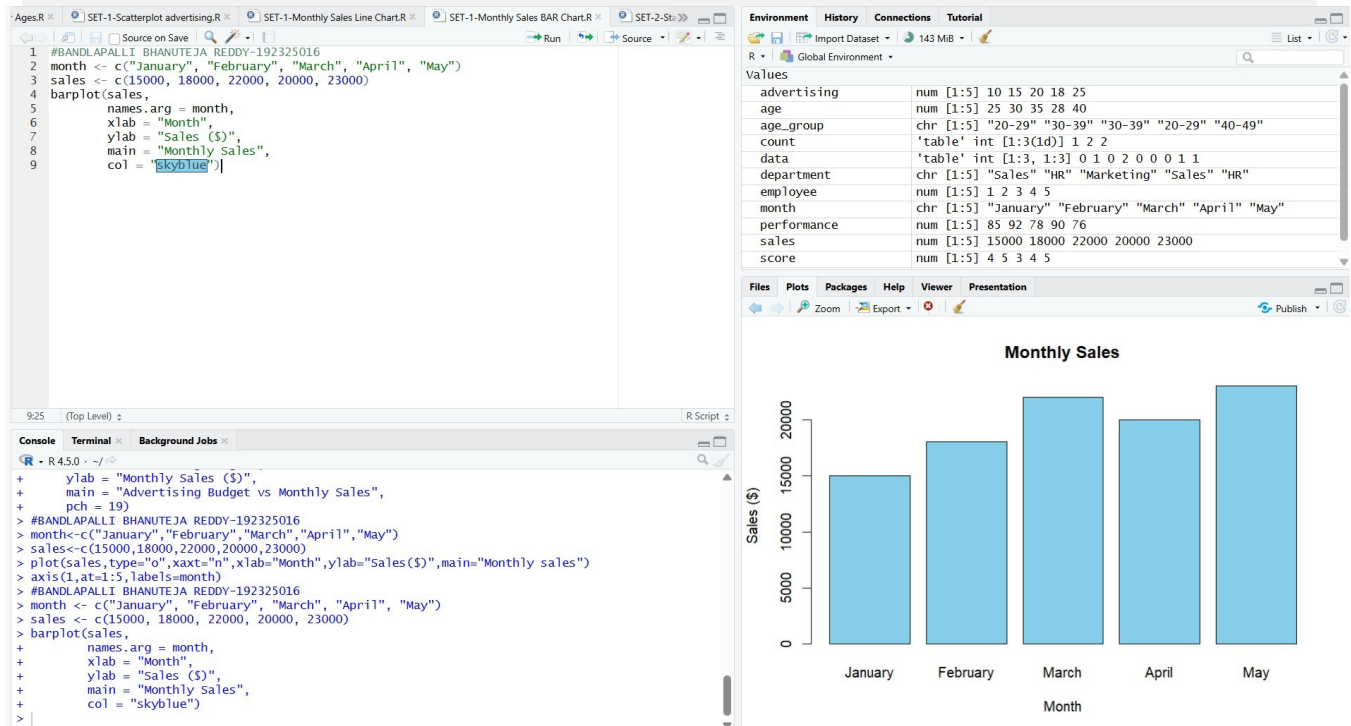
Observation: Sales increase from January to March, decrease slightly in April, and reach the highest value in May.

PROGRAM 2: Bar Chart of Sales

Generate a bar chart to display the sales values for the available months and label the chart elements.

```
# Dataset month <- c("January", "February", "March",
"April", "May") sales <- c(15000, 18000, 22000, 20000,
23000)
```

```
# Bar Chart barplot(sales,
names.arg = month,      xlab
= "Month",              ylab =
"Sales ($)",            main =
"Monthly Sales",        col =
"skyblue")
```



Output 2: Monthly Sales Bar Chart

Observation: The bar chart makes it easy to compare the sales of each month. May has the highest sales and January has the lowest.

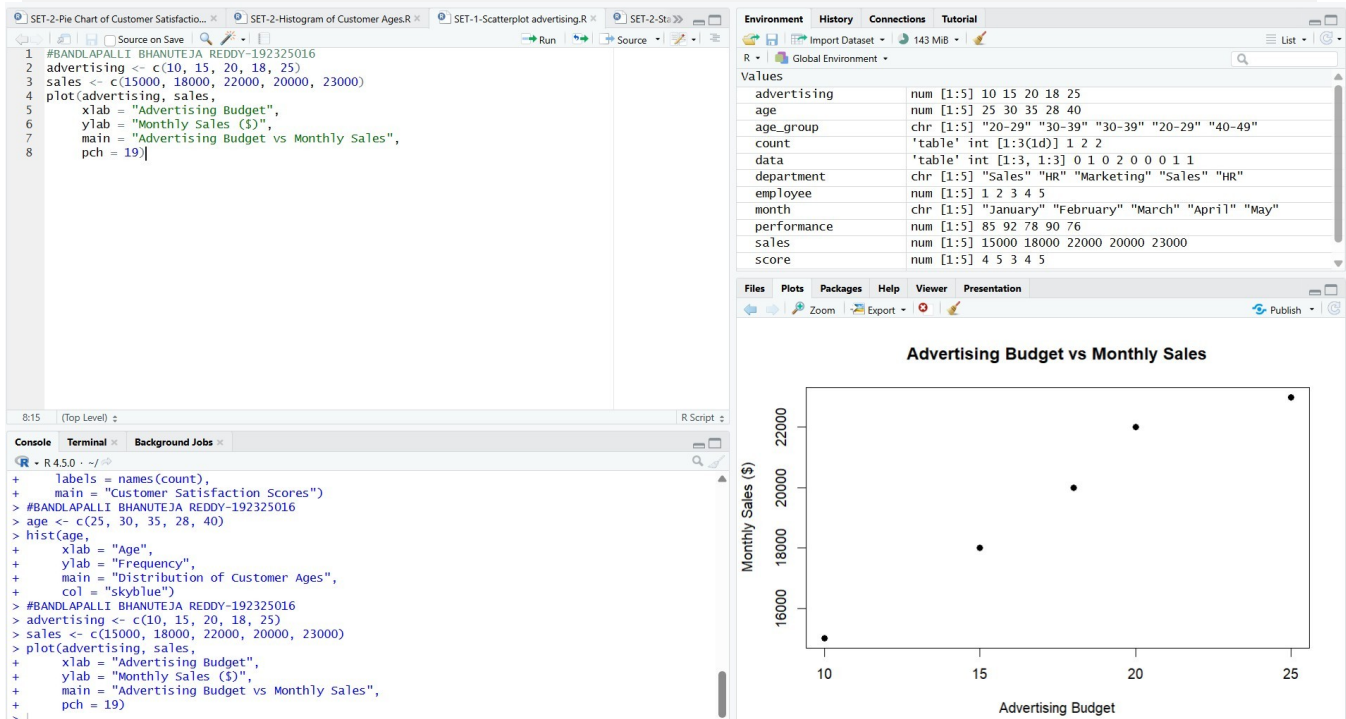
PROGRAM 3: Scatter Plot of Advertising Budget vs Monthly Sales

Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

```
# Dataset sales <- c(15000, 18000, 22000,
20000, 23000)
```

```
advertising <- c(5000, 6000, 7000, 8000, 9000)
```

```
# Scatter Plot plot(advertising, sales, main =
"Advertising Budget vs Monthly Sales", xlab =
"Advertising Budget ($)", ylab = "Monthly
Sales ($)", pch = 19, col = "red")
```



Output 3: Advertising Budget vs Monthly Sales Scatter Plot

Insight: The plotted points show a generally positive relationship: as the advertising budget increases from \$5,000 to \$9,000, monthly sales also generally increase. This suggests that higher advertising spending is associated with higher sales for the values plotted.

Note: The visible assignment table contains only Month and Sales. The advertising values used in the submitted R program are the five values shown in the user's executed program: 5000, 6000, 7000, 8000, and 9000.

SET 2: CUSTOMER FEEDBACK ANALYSIS

Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
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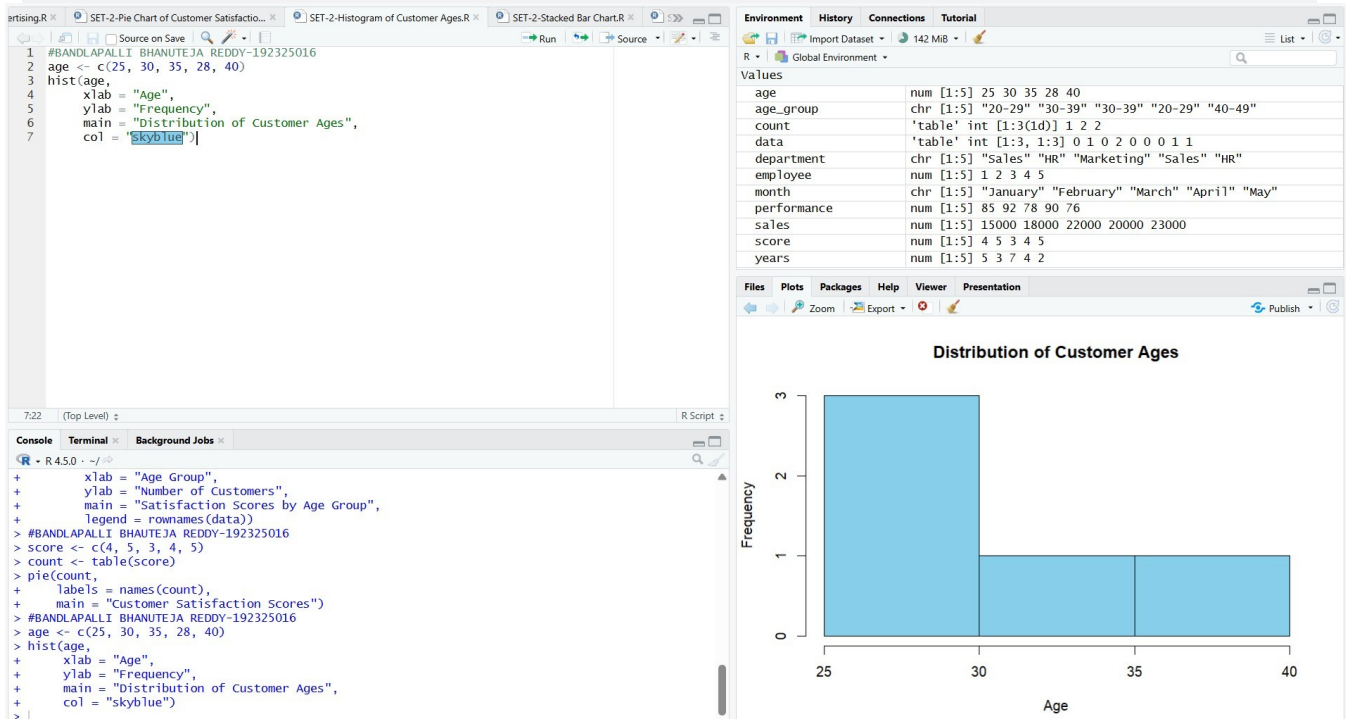
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

PROGRAM 1: Histogram of Customer Ages

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
# Dataset age <- c(25, 30,
35, 28, 40)

# Histogram hist(age,    main =
"Distribution of Customer Ages",    xlab =
"Age",    ylab = "Frequency",    col =
"skyblue")
```



Output 1: Distribution of Customer Ages

Observation: The histogram represents the distribution of the five customer ages in the given dataset.

PROGRAM 2: Pie Chart of Satisfaction Scores

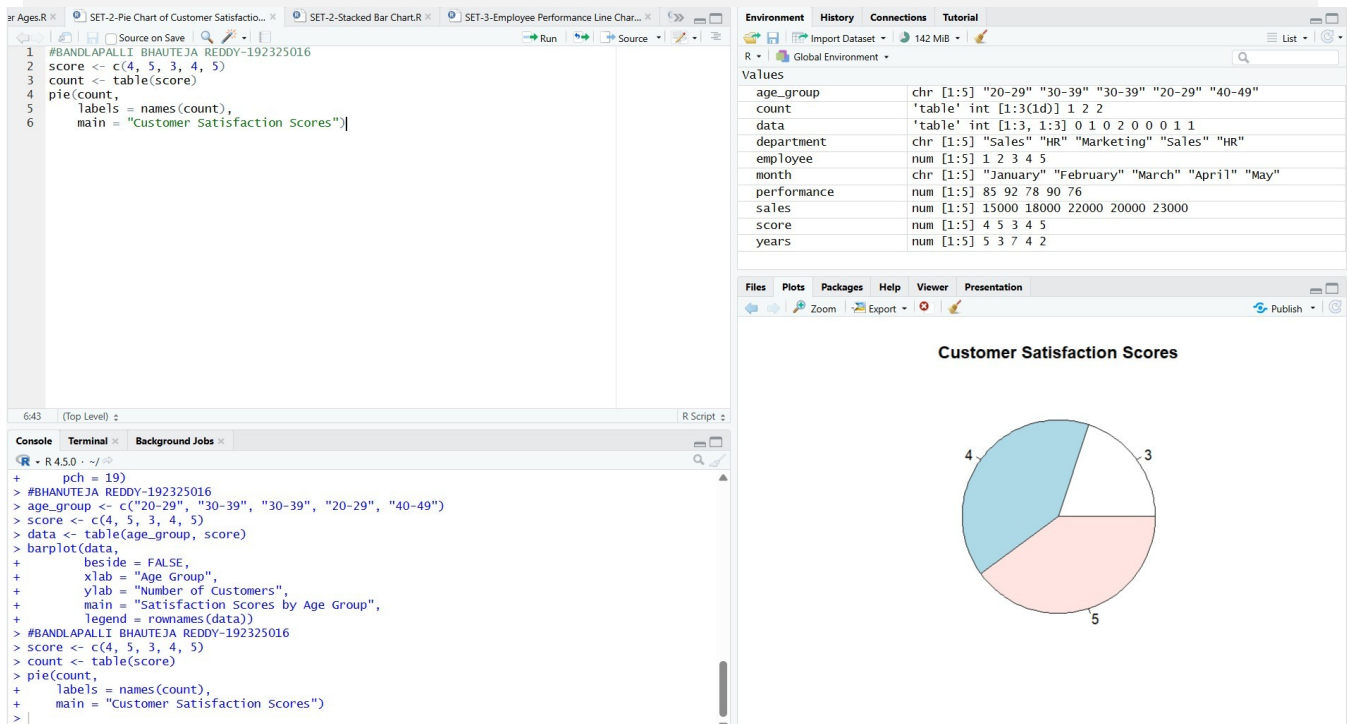
Generate a pie chart to display the overall distribution of customer satisfaction scores and include labels.

```
# Satisfaction scores
```

```
satisfaction <- c(4, 5, 3, 4, 5)
```

```
# Calculate frequency score_count
<- table(satisfaction)
```

```
# Pie Chart pie(score_count, labels =
paste(names(score_count), "Score"), main =
"Customer Satisfaction Scores")
```



Output 2: Customer Satisfaction Scores Pie Chart

Observation: Scores 4 and 5 each occur twice, while score 3 occurs once.

PROGRAM 3: Stacked Bar Chart by Age Group

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

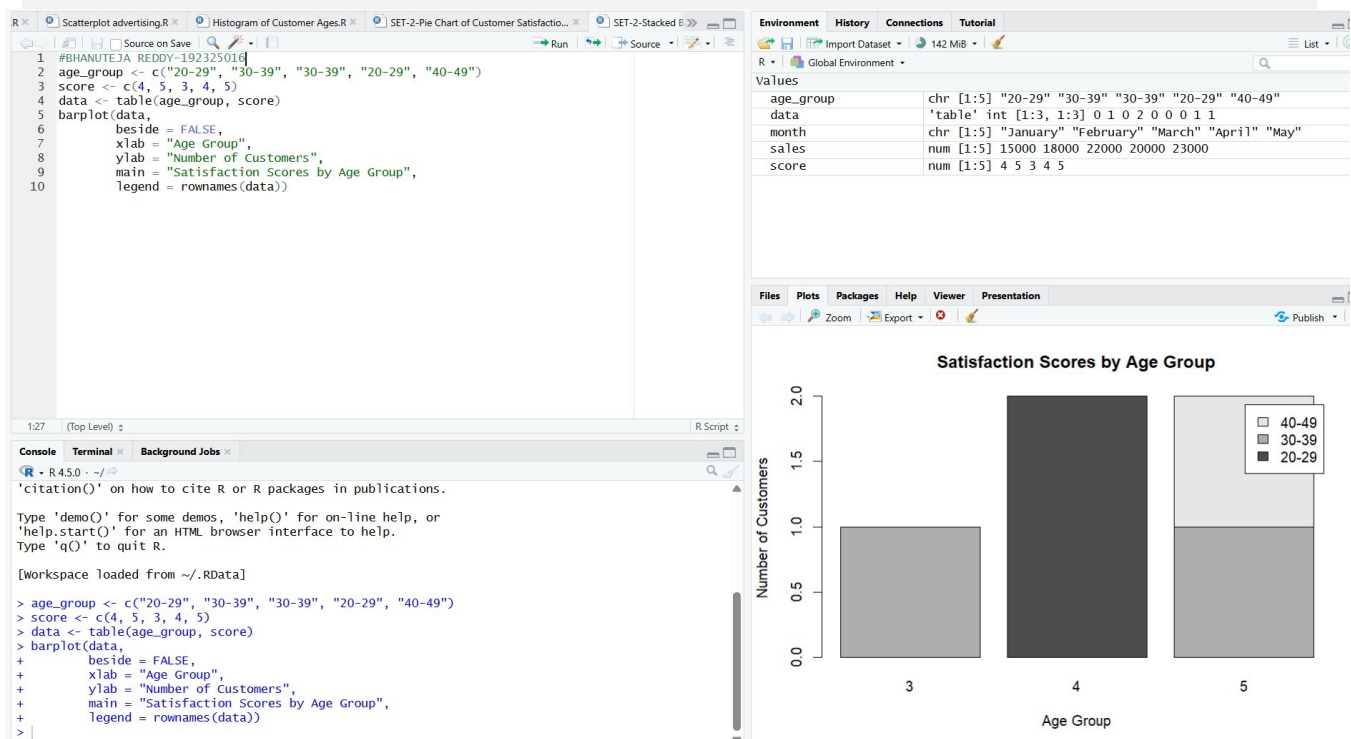
```
# Dataset age <- c(25, 30,
35, 28, 40) satisfaction <-
```

```
c(4, 5, 3, 4, 5)
```

```
# Create age groups age_group
<- cut(age,
      breaks = c(20, 29, 39, 49),
      labels = c("20-29", "30-39", "40-49"))
```

```
# Create table data <-
table(age_group, satisfaction)
```

```
# Stacked Bar Chart barplot(data, beside =
FALSE, main = "Satisfaction Scores by Age
Group", xlab = "Age Group", ylab =
"Number of Customers", col = c("skyblue",
"lightgreen", "orange"), legend.text =
TRUE)
```



Output 3: Satisfaction Scores by Age Group

Observation: The code creates three age groups and stacks the satisfaction-score frequencies for each group.

Note: The supplied RStudio screenshot is inserted exactly as captured, including its displayed x-axis labels.

SET 3: EMPLOYEE PERFORMANCE EVALUATION**Dataset: Employee Performance**

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

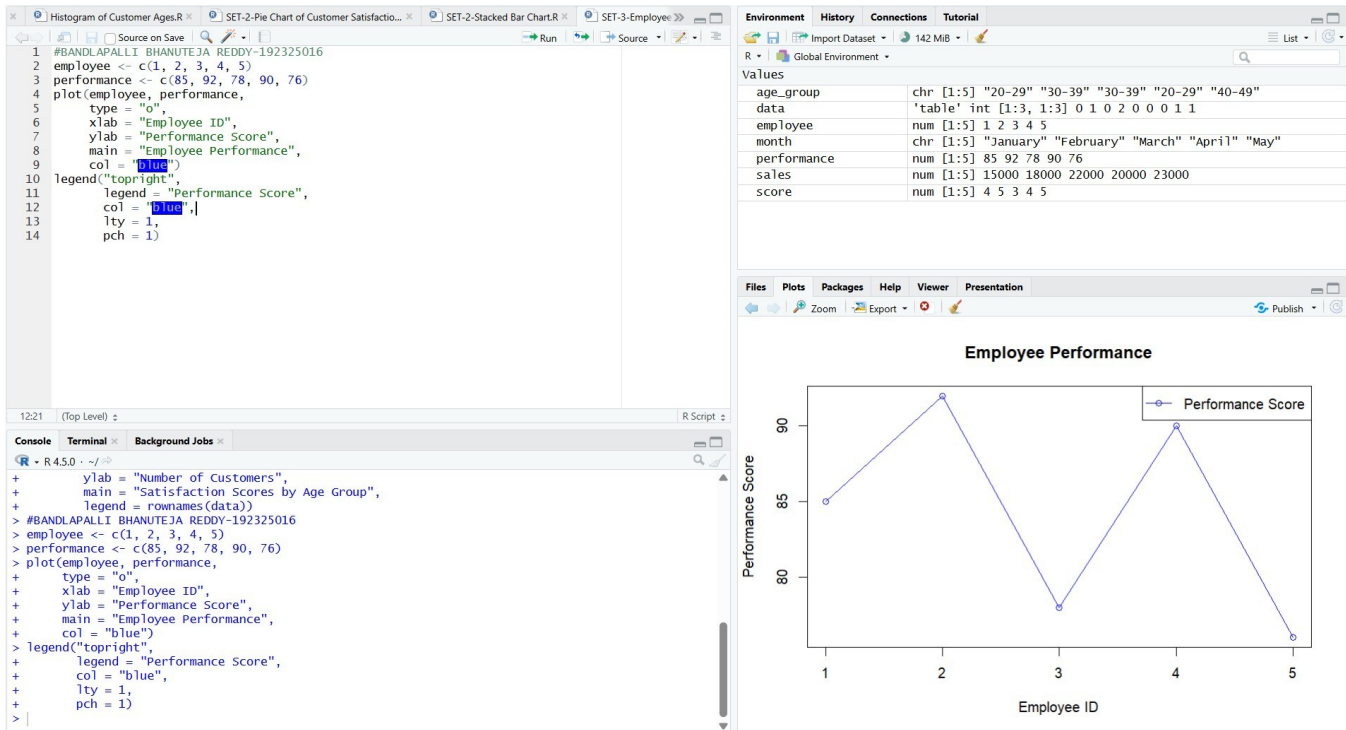
PROGRAM 1: Line Chart of Performance Trend

Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

```
# Dataset employee_id <- c(1, 2, 3,
4, 5) performance <- c(85, 92, 78,
90, 76)

# Line Chart plot(employee_id,
performance,    type = "o",    pch = 19,
col = "blue",    ylim = c(70, 100),    xlab
= "Employee ID",    ylab = "Performance
Score",    main = "Employee Performance
Trend")

# Legend legend("topright",
legend = "Performance Score",
col = "blue",    lty = 1,    pch
= 19)
```



Output 1: Employee Performance Trend

Observation: Performance rises from 85 (Employee 1) to a peak of 92 (Employee 2), falls to 78 for Employee 3, recovers to 90 for Employee 4, and drops to the lowest value of 76 for Employee 5. The legend identifies the plotted series.

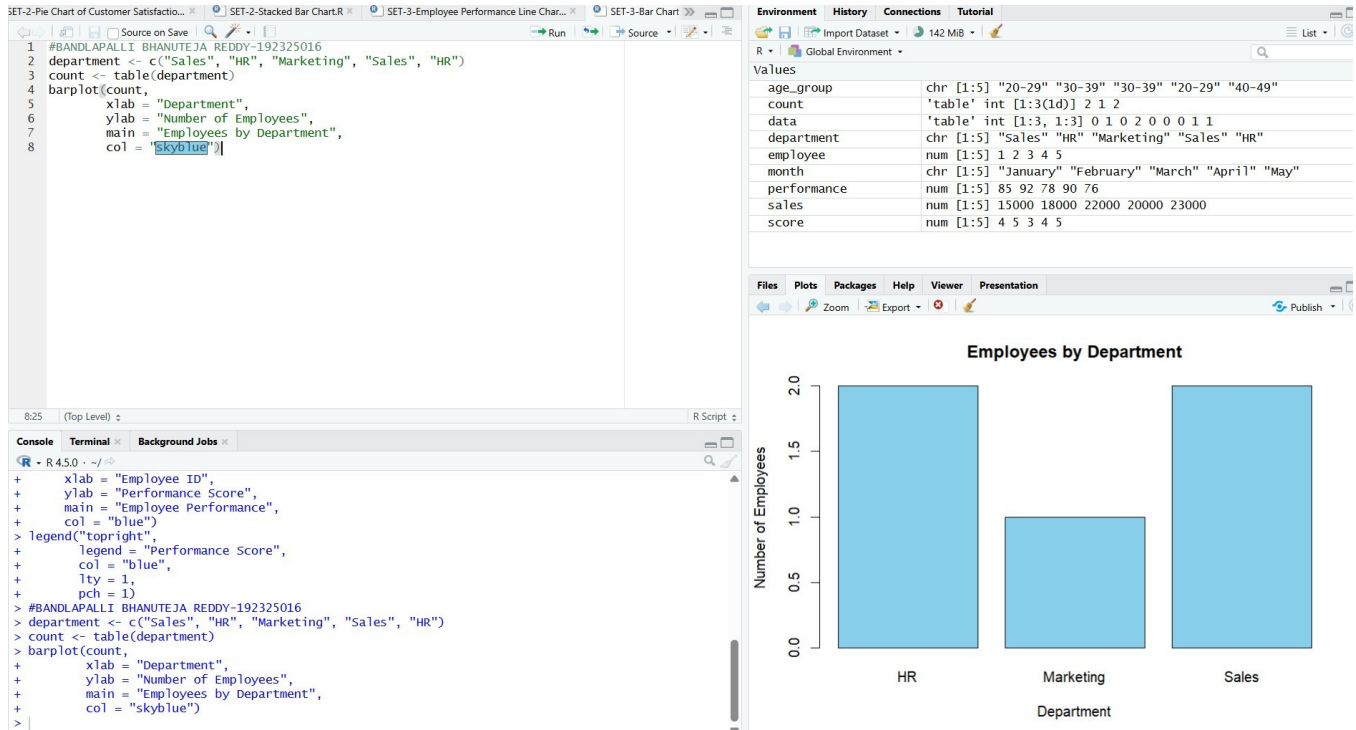
PROGRAM 2: Bar Chart of Employees by Department

Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
# Dataset department <- c("Sales", "HR", "Marketing",
"Sales", "HR")
```

```
# Count employees in each department
dept_count <- table(department)
```

```
# Bar Chart barplot(dept_count, xlab =
"Department", ylab = "Number of
Employees", main = "Employees by
Department", col = c("skyblue",
"lightgreen", "orange"))
```



Output 2: Employees by Department

Observation: Sales and HR each have 2 employees, while Marketing has 1 employee. Sales and HR together account for 4 of the 5 employees.

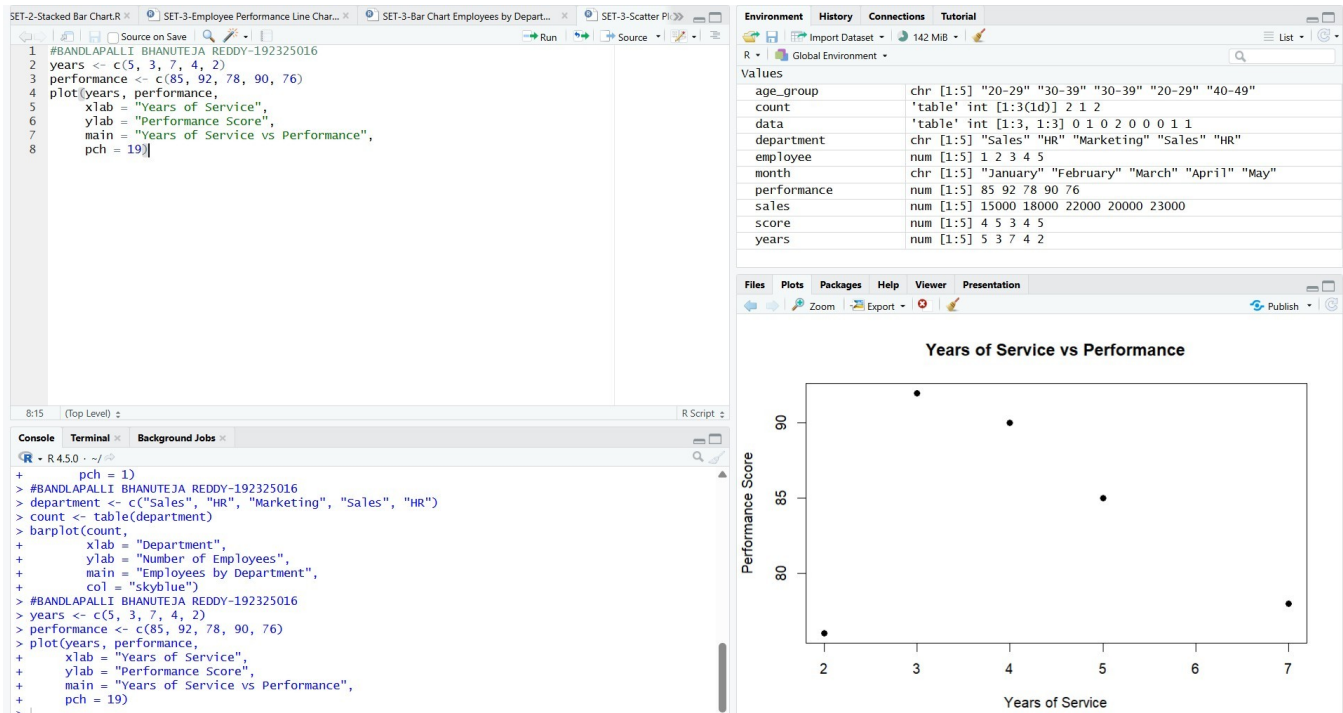
PROGRAM 3: Scatter Plot of Years of Service vs Performance

Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

```
# Dataset years <- c(5, 3, 7, 4, 2)
performance <- c(85, 92, 78, 90, 76)

# Scatter Plot plot(years, performance, pch = 19,
col = "red", xlab = "Years of Service", ylab =
"Performance Score", main = "Years of Service
vs Performance Score")

# Trend line and correlation abline(lm(performance
~ years), col = "blue") cor(years, performance)
```



Output 3: Years of Service vs Performance Score

Insight: The correlation coefficient is about -0.17, which indicates a very weak negative relationship. The points are scattered widely around the trend line, so years of service does not clearly predict performance. For example, Employee 3 (7 years) scored 78, while Employee 4 (4 years) scored 90.

Note: Only five records are available, so this result is indicative and should not be generalized.